

Opinion

# The feasibility of artificial consciousness through the lens of neuroscience

Jaan Aru,<sup>1,\*</sup> Matthew E. Larkum,<sup>2,\*</sup> and James M. Shine<sup>3,\*</sup>

Interactions with large language models (LLMs) have led to the suggestion that these models may soon be conscious. From the perspective of neuroscience, this position is difficult to defend. For one, the inputs to LLMs lack the embodied, embedded information content characteristic of our sensory contact with the world around us. Secondly, the architectures of present-day artificial intelligence algorithms are missing key features of the thalamocortical system that have been linked to conscious awareness in mammals. Finally, the evolutionary and developmental trajectories that led to the emergence of living conscious organisms arguably have no parallels in artificial systems as envisioned today. The existence of living organisms depends on their actions and their survival is intricately linked to multi-level cellular, inter-cellular, and organismal processes culminating in agency and consciousness.

## Large language models and consciousness

There is a long tradition of questioning which animals are conscious [1–3] and whether entities outside the animal kingdom might be conscious [4–6]. Recently, the advent of LLMs has brought a novel set of perspectives to this question. Through their competence and ability to converse with us, which in humans is indicative of being conscious, LLMs prompt us to refine current notions of what it means to understand, to have agency, and to be conscious.

LLMs are sophisticated, multi-layer artificial neural networks with billions of connections whose weights are trained on hundreds of billions of words from various texts, including natural language conversations between humans. Through text-based queries, users interacting with LLMs are provided with a fascinating language-based simulation. If you take the time to use these systems, it is hard not to be struck by the apparent depth and quality of the internal machinations in the network. Ask it a question and it will provide you with an answer that drips with the kinds of nuance we typically associate with conscious thought. As a discerning, conscious agent yourself, it is tempting to conclude that the response has been generated by a similarly conscious being, one that thinks, feels, reasons, and experiences. Using this type of a ‘Turing test’ as a benchmark, the question can be raised whether LLMs are or soon will be conscious [7–10], which in turn raises a host of moral quandaries, such as whether it is ethical to continue to develop LLMs that could be on the brink of conscious awareness. While this position might not be prevalent among neuroscience researchers today, the improving capabilities of artificial intelligence (AI) systems will inevitably lead to the point where the possibility of machine consciousness needs to be addressed. Furthermore, this possibility is discussed extensively in news media, prompting neuroscientists to consider some of the arguments in favor and against it.

The notion of LLMs’ potential to be conscious is often bolstered by the fact that the architecture of LLMs is loosely inspired by features of brains (Figure 1), the only objects to which we can currently attribute consciousness with confidence. However, while early generations of artificial neural

## Highlights

Large language models (LLMs) can produce text that leaves the impression that one may be interacting with a conscious agent.

Present-day LLMs are text-centric, whereas the phenomenological Umwelt of living organisms is multifaceted and integrated.

Many theories of the neural basis of consciousness assign a central role to thalamocortical re-entrant processing. Currently, such processes are not implemented in LLMs.

The organizational complexity of living systems has no parallel in present-day AI tools. Possibly, AI systems would have to capture this biological complexity to be considered conscious.

LLMs and the current debates on conscious machines provide an opportunity to re-examine some core ideas of the science of consciousness.

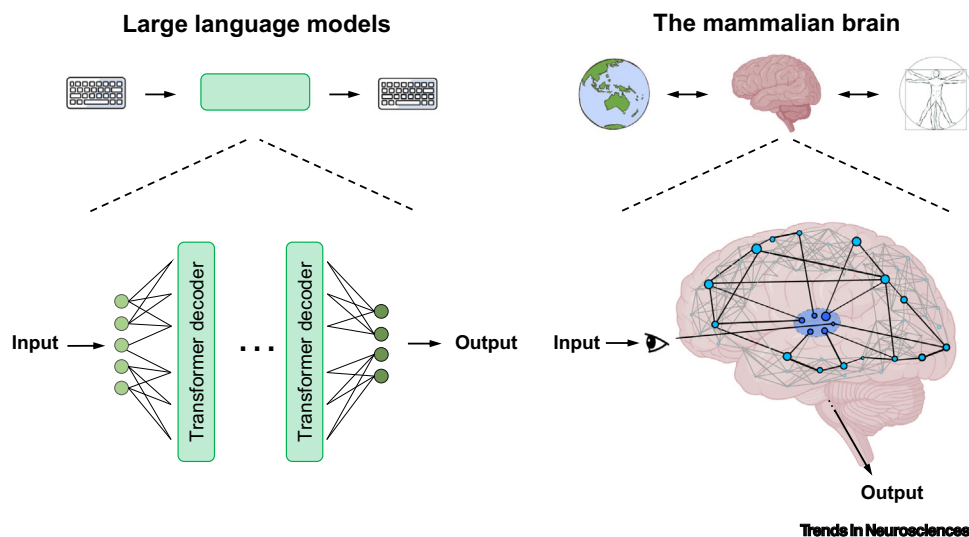
<sup>1</sup>Institute of Computer Science, University of Tartu, Tartu, Estonia

<sup>2</sup>Institute of Biology, Humboldt University Berlin, Berlin, Germany

<sup>3</sup>Brain and Mind Center, The University of Sydney, Sydney, Australia

\*Correspondence: jaan.aru@ut.ee (J. Aru), larkumma@hu-berlin.de (M.E. Larkum), and mac.shine@sydney.edu.au (J.M. Shine).





**Figure 1. Macroscopic topological differences between mammalian brains and large language models.** Left: a schematic depicting the basic architecture of a large language model, which can have tens or even more than a hundred decoder blocks arranged in a feed-forward fashion. The inputs and outputs are strings of letters. Right: a heuristic map of the thalamocortical system, which generates complex neural activity patterns thought to underlie consciousness. Input is multimodal information from the world and output is created by the interaction between the brain and the body.

networks were designed as a simplified version of the cerebral cortex [11], modern LLMs have been highly engineered and fit to purpose in ways that do not retain deep homology with the known structure of the brain. Indeed, many of the circuit features that render LLMs computationally powerful (Figure 1) have strikingly different architectures from the systems to which we currently ascribe causal power in the production and shaping of consciousness in mammals. For instance, many theories of the neural basis of consciousness would assign a central role in conscious processing to thalamocortical [12–17] and arousal systems [18–24], both features that are architecturally lacking in LLMs.

One might ask why it is so crucial for the architecture of LLMs to mimic features of the brain. The primary reason, in our view, is that we can currently be absolutely sure of only a version of consciousness that arises from brains embedded within complex bodies. Some may contend that in its strictest form, this argument could be further collapsed to humans, though many of the systems-level features considered important for subjective consciousness are pervasive across phylogeny, stretching back to mammals [13,24,25] and even to invertebrates [26]. We will return to this point, but start with the question about what precisely we mean by the term ‘consciousness’. From there we will develop three arguments against the view that present-day AI systems have, or that future AI systems will soon have, consciousness: first, consciousness is tied to the sensory streams that are meaningful for the organism; second, in mammalian brains, consciousness is supported by a highly interconnected thalamocortical system; and third, consciousness might be inextricably linked to the complex biological organization characteristic of living systems.

### What is consciousness?

Consciousness is a complex concept and its definitions have long been debated. In the context of human interactions, conversation would be among the first elements typically used to assess whether another person is conscious or not. As discussed earlier, interactive language-based conversations with LLMs are currently often a starting point to develop an intuitive sense about

whether LLMs might be conscious. Although these conversations are remarkable, they are not formal objective measures of consciousness and constitute only *prima facie* evidence for conscious agency. The advent of LLMs has demanded a re-evaluation of whether one can indeed infer consciousness directly from verbal interactions with other agents. Thus, there is an emerging view that the criteria for attributing human-like abilities and characteristics need to be re-assessed [27].

There are different meanings associated with the word ‘consciousness’. Neurologists, for instance, often refer to *levels* of consciousness; in the first place, whether a person is conscious or not and, in a more refined manner, assessing the gradations or specific states of consciousness. Psychologists, by contrast, often focus on the contents of consciousness: the specific experiences, memories, and thoughts of individuals’ inner world. Furthermore, there are distinctions between different contents of consciousness: our experiences can be described, for instance, as primarily phenomenal or experiential [28] (e.g., the sight/smell of an apple, or the feel of your arm) or more abstract [28] (e.g., how we imagine, prospect, or manipulate concepts). The question of whether AI systems are conscious could be approached in various ways: it could focus primarily on only some of these aspects, or possibly all of them together. In the following, we focus mainly on phenomenal consciousness and ask whether machines can experience the world phenomenally.

### The umwelt of an LLM

The aspect of the world that is perceptually ‘available’ to an organism has been described as its ‘*umwelt*’ (from the German ‘environment’ [29]). For instance, human retinas respond to wavelengths of light ranging from ~380 to 740 nm, which we perceive as a spectrum from blue to red. Without technological augmentation, we cannot detect light waves outside of this narrow band, in the infrared (>740 nm) or UV (<380 nm) bands. We have a similar *umwelt* for the auditory domain (we cannot hear tones outside the 20–20 000 Hz range), somatosensory domain (we can differentiate stimulation up to about 1 mm apart on some parts of our body), and vestibular domain (yoked to the 3D structure of our semicircular canals, which provide our inner sense of ‘balance’). Other species can detect other portions of the electromagnetic spectrum. For instance, honeybees can see light in the UV range [30] and some snakes can detect infrared radiation in addition to more traditional visual light cues [31]; that is, the bodies and brains of other animals place different constraints on their sensitivity to the sensory world around them. Gibson referred to this information, that we can pragmatically interact with, as a set of ‘affordances’ [25,32–34].

If anything at all, what is the *umwelt* of an LLM? What kinds of affordances does an LLM have access to? By the nature of its design, an LLM is only ever presented with binary-coded patterns fed to the network algorithms inherent within the complex transformer architectures that comprise the inner workings of present-day LLMs [35,36]. While neuronal spikes also potentially encode incoming analog signals as digital (i.e., binary), the information stream fed to the LLMs in question is highly abstract and hence does not itself make any robust contact with the world as it is. Text and speech coded into strings of letters are simply no match for the dynamic complexity of the natural world: the *umwelt* of an LLM (the information afforded to it) is of a fundamentally different nature compared with the information that enters our brain when we open our eyes or listen to a conversation, and hence any accompanying experience. Traditional philosophical discourse has underscored the distinctiveness in the information streams experienced across species (for instance, between humans and bats [37]) and the phenomenology of these experiences. While there is no definite way to quantify this difference, we highlight that the informational input accessible to LLMs is likely to exhibit a more significant disparity.

That being said, it is worth mentioning that there is no conceptual barrier stopping the input of future AI systems from being much more enriched. Future LLMs could be equipped with different types of inputs (see [38,39]) that better match the kinds of signals that conscious agents have access to every day (i.e., the statistics of the natural world). Taking this even further, could the *umwelt* of future AI systems become more extended than that available to humans? In contemplating this question, it is essential to recognize that our *umwelt* and conscious experience are not determined solely by sensory input. For example, consider lying in a floatation tank where, despite a lack of normal sensory experiences, consciousness persists. This underscores the notion that having an *umwelt* presupposes an inherent subjective perspective, that is, an agent to begin with [29,40,41]. Similarly, affordances depend on the internal properties of the agents, in particular their motivations and goals [33,40,41]. This underscores the point that consciousness does not arise merely from data and hence that simply adding massive data streams to future AI systems will not, by itself, lead to consciousness.

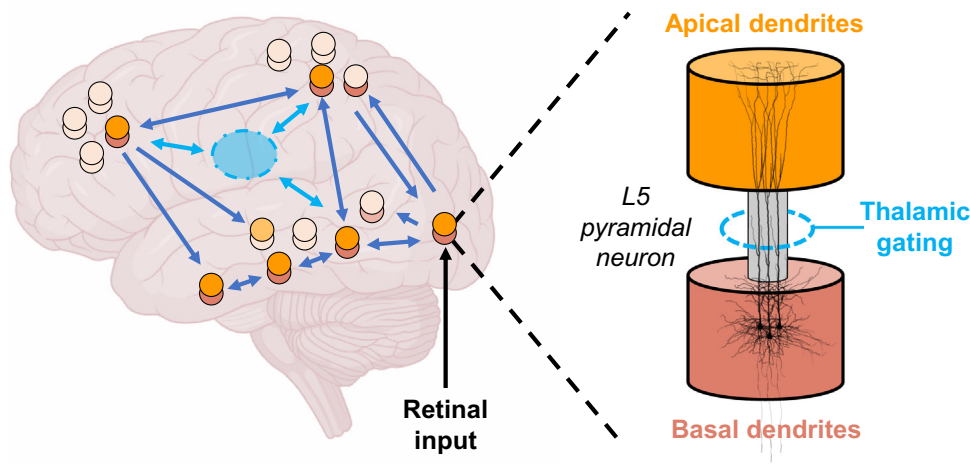
These notions may prompt us to rethink some of the fundamental assumptions in the science of consciousness. Specifically, as AI systems continue to exhibit increasingly sophisticated abilities, one will have to re-evaluate the necessity of more basic self- and agency-related processes for the emergence of consciousness, as posited by some of the theories of consciousness [42–46].

### The neural architecture supporting conscious integration

There is a sizable literature on the neural correlates of consciousness, with many different theories about the neural processes that underlie conscious processing. Some of these frameworks highlight that consciousness is supported by neural processing within the dense, re-entrant thalamocortical network [12–15,47–53]. The thalamocortical network encompasses cortical areas, cortico-cortical connectivity, and higher-order thalamic nuclei with their diffuse projections to cortical areas [54–56]. This specific architecture of the thalamocortical system supports recurrent and complex processing thought to underlie consciousness [53,57–61] and conscious integration (i.e., the fact that consciousness feels unified despite arising from processes happening in different brain areas) [51,53,62]. However, the details of how this integration is achieved differ across various theories of consciousness.

For instance, according to the global neuronal workspace theory [48,49] consciousness depends on the central workspace constituted by a distributed frontoparietal cortical system. This workspace integrates information from local cortical processors and then globally broadcasts it to all local processors, with the global broadcast delineating conscious from non-conscious processes. Other theories of consciousness assign a different neural process to carry out this integration. For instance, the binding-by-synchrony theory [63,64] suggests that conscious integration occurs via high-frequency synchronization between different cortical areas, a process that can be putatively involved in diverse functions, including perception, cognition, or motor planning, depending on the cortical regions involved.

In the dendritic integration theory (DIT) [12,53] (Figure 2), it is proposed that global conscious integration also depends on local integration at the level of single layer 5 pyramidal neurons, which are large excitatory neurons that hold a central position in both thalamocortical and cortico-cortical loops [12,53]. These neurons have two major compartments (Figure 2, orange and red cylinders) that process categorically distinct types of information: the basal compartment (red) processes externally-grounded information, whereas the apical compartment (orange) processes internally-generated information [12,53,65]. According to the DIT, during conscious states, these two compartments are coupled (i.e., integrated), allowing information to flow



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**Figure 2.** The neural architecture underlying conscious integration according to the dendritic integration theory (DIT). Left: sensory inputs (for instance, visual inputs) contain information that drives feed-forward activity in the sensory system. Internally generated signals (e.g., prediction, memory, attention) can augment certain features of the input stream (represented in the figure by bilateral arrows). This makes these representations stand out from the background (orange/red), leaving others inactive (light red). According to DIT [12,53], the thalamus (light blue; dot-dashed line) plays a crucial role in shaping/gating the contents of consciousness. Right: DIT assigns a key role to the subset of thick-tufted layer 5 (L5) pyramidal neurons in consciousness. These neurons display burst-firing, which occurs when depolarization of the cell body via basal dendrites (red) coincides temporally with descending cortical inputs to apical dendrites (orange), particularly in the presence of gating inputs from higher-order, matrix-type thalamus (blue).

through the thalamocortical and cortico-cortical connections, thus enabling system-wide integration and consciousness [12,53].

Notably, the architectures of present-day LLMs and other AI systems are devoid of features from each of these theoretical proposals: there is no equivalent of dual-compartment pyramidal neurons, nor a centralized thalamic architecture, a global workspace, or the many arms of the ascending arousal system. In other words, these AI systems are missing the very features of brains that are currently hypothesized to support consciousness. Although we are not arguing that the mammalian brain is the only architecture capable of supporting conscious awareness, the evidence from neurobiology suggests that very specific architectural principles (i.e., more than simple connections between integrate-and-fire neurons) are responsible for mammalian consciousness (see [1–4,26] for some examples of research on consciousness in non-mammalian species). Topologically, present-day AI systems are extremely simple in comparison, which is among the reasons we are cautious in ascribing phenomenal consciousness to them.

Could future AI models eventually incorporate the process of ‘integration’, which many theories of consciousness see as central? The integration proposed by the global neuronal workspace theory [48,49] offers a relatively straightforward implementation [9,10] and, in fact, some recent AI systems have incorporated something akin to a global workspace shared by local processors [66,67]. As the computational process of global broadcasting can be implemented in AI systems, an artificial system with a computationally equivalent global workspace would include a core ingredient underlying consciousness according to this theory [9,10]. However, as indicated earlier, not all theories of consciousness agree that this type of integration is key to consciousness. For instance, the integrated information theory of consciousness [50,51] claims that it is

impossible for a software-based AI system instantiated on a typical modern computer to achieve consciousness because modern computers do not have the appropriate architecture to realise the cause-effect power necessary for sufficiently integrating information [68,69]. Here, we will consider a third possibility, namely that consciousness might be implementable in principle, but it might require a level of computational specificity that is beyond the present-day (and perhaps future) AI systems.

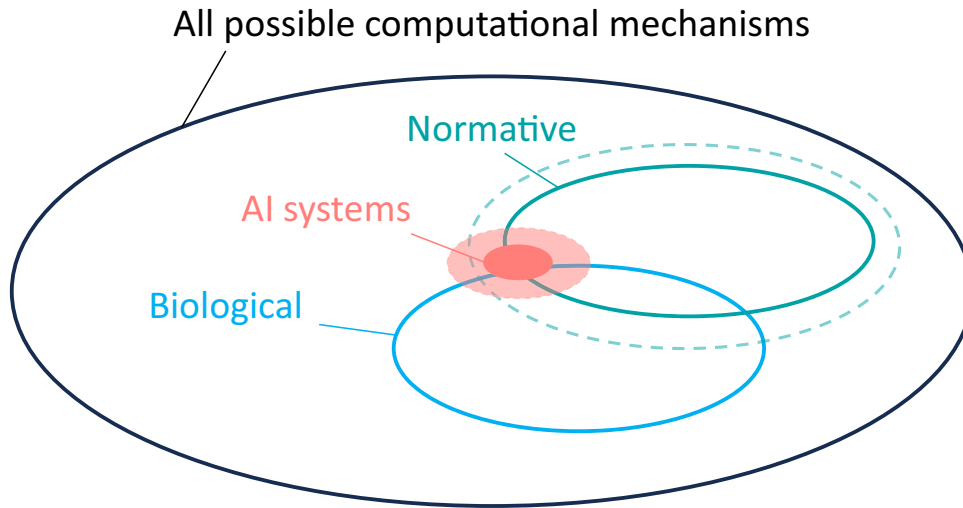
### Consciousness as a complex biological process

Consciousness does not only depend on the architecture of the system. For instance, the structure of the thalamocortical system does not change when we are in deep sleep or undergo anesthesia, yet consciousness disappears. Even during deep sleep, local neural responses and gamma-band activity in primary sensory areas can remain similar to those in the conscious state [70,71]. This implies that consciousness relies on specific neural processes that are different in conscious and unconscious brains.

To illustrate current knowledge about the details distinguishing conscious from unconscious processing, we will return to DIT, which encapsulates some of the specific neurobiological nuances relevant to the matter. In particular, DIT proposes that the crucial difference between conscious and unconscious processing lies in the integration between the two compartments of pyramidal cells (Figure 2). As indicated earlier, during conscious processing, these two compartments interact and hence enable complex and integrated processing across the thalamocortical system [12,53]. By contrast, it has been demonstrated that various anesthetic agents lead to functional decoupling between the two compartments [72]. In other words, while anatomically the neurons are intact and can fire action potentials, physiologically dendritic integration is severely limited within these cells: the top-down feedback component cannot influence processing. This dendritic coupling was demonstrated to be controlled by metabotropic receptors, which are often overlooked in computational models and in artificial neural networks. Furthermore, it was suggested that the activity of the metabotropic receptors in this context might be controlled by the higher-order thalamus [72]. Thus, there are relatively specific neurobiological processes that may be responsible for switching consciousness 'on' and 'off' in the brain. Clearly, the quality of experience in mammalian brains is intricately linked to the underlying processes from which it arises.

However compelling, these details almost certainly pale compared with the true complexity of the depth of biological organization required to achieve a satisfactory understanding of consciousness. While today's explanations of consciousness rely on ideas such as the global workspace, information integration, recurrent feedback, dendritic integration, and other notions, it might be the case that the biological processes underlying consciousness are more intricate than these current concepts appreciate. It is also quite possible that the abstract computational-level ideas that are currently used to frame discussions in consciousness research may miss the necessary computational details required to satisfactorily explain consciousness. In other words, biology is complex and our understanding of biological computations is limited (Figure 3), so perhaps we simply do not yet have the right mathematical and experimental tools to understand consciousness.

To better ground this notion of biological complexity, it is worth considering that the cellular- and system-level processes described earlier are inextricably embedded within a living organism. Living organisms differ from present-day machines and AI algorithms, as they are constantly in the process of self-maintenance across several levels of processing [73,74]. Also, living systems have a multifaceted evolutionary and developmental history and their existence depends on their



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**Figure 3. The limitations of current computational understanding of consciousness.** The space (in the mathematical, abstract sense) of all possible computations (schematized as the large ellipse in the figure) is broader than the types of ‘normative’ computations currently envisioned or formalized in computational models (teal ellipse). Hence, current conceptualization may not have yet captured the key computations underlying consciousness. It can be argued that some of the normative computations that comprise the biological processes responsible for consciousness are currently understood (overlap between teal and blue ellipse). However, the ‘knowledge horizon’ of computational mechanisms is evolving and perhaps would have to be further extended (broken teal ellipse) for understanding consciousness. Computations in artificial intelligence (AI) systems (red ellipse) show some overlap with biological computations and some of the computations of AI systems are understood. However, the computations of AI systems differ from those of biological systems. In view of these differences, there is little *a priori* reason to assume, we would argue, that the computations of present-day AI systems are related to computations underlying phenomenal consciousness.

actions on multiple levels of organization (i.e., they have ‘skin in the game’; [Box 1](#)). It has been argued that consciousness is intricately linked with the organization of living systems [74–77]. Here, we would like to draw attention to the fact that this organizational complexity (i.e., interactions between the different levels of the system) [77–81] of living systems is not captured within present-day computer software. While this fact need not impede progress in AI, it is entirely possible that the lack of any constraints imposed on modern AI algorithms to work like a living system effectively means that as long as AI is based on software, AI might be poorly placed to recapitulate conscious experience and agency.

The notion of biological complexity outlined in the previous paragraph is relevant also at the cellular level. A biological neuron is not just an abstract entity that can be fully captured with a few lines of code. Biological cells have multi-level organization and depend on a further cascade of biophysical intracellular complexity [79–83]. Consider the Krebs cycle, for instance, which underlies cellular respiration, a key process in maintaining cellular homeostasis [84]. Cellular respiration is a crucial biological process that enables cells to convert the energy stored in organic molecules into a form of energy that can be utilized by the cell. This process, however, is not ‘compressible’ into software, as processes like cellular respiration need to happen with real physical molecules. To be clear, our aim is not to suggest that consciousness requires the Krebs cycle, but rather to highlight that understanding consciousness may involve similar challenges in translating from the biological to the artificial realms: perhaps it cannot be abstracted away from the underlying machinery [68,69,85].

### Box 1. LLMs and skin in the game: do we have a moral quandary?

Does it really matter if LLMs are conscious? If LLMs can match and even exceed our expectations in terms of getting superficially human-like responses that are useful and informative, is there any need to speculate about what an LLM experiences? Some argue that from a moral perspective, the answer should be a definitive 'yes' [92]. According to this view, we should carefully consider the ethical implications for any conscious entity, including AI, principally because it is assumed that some experiences could be negative and that in this case the AI system could also suffer. At this point, it is claimed, we should care or at least be cautious about whether an AI system might suffer.

We claim that LLMs do not (and will not) have experiences that can be considered suffering in any sense that should matter to human society. The notion of 'skin in the game' [93] may be useful as an analogy for articulating our argument here. This notion emphasizes the importance of personal investment and engagement in moral decision-making and suggests that those who have a personal stake in an issue are more competent to make ethical judgments than those who do not [93]. An LLM could, in principle, state in a conversation that it does not want to be shut down, but an LLM does not have 'skin in the game', as arguably there is no real consequence to the software when it is actually shut down. In contrast, in biology, the system has something to lose on several levels [73]: if it stops living, it will die. As the philosopher Hans Jonas has said: 'The organism has to keep going, because to be going is its very existence' [94]. If cellular respiration stops, the cell may die; if cells die, organs fail; if organs fail, the organism will soon die. The system has skin in the game across levels of processing, which is arguably prerequisite for caring about agency and consciousness [73]. Here, we would argue that not having the capacity for phenomenal consciousness would preclude suffering and, therefore, personal investment.

Importantly, we are not necessarily subscribing to the claim that consciousness cannot be captured within software at all [68,69,85–87]. Rather, we emphasize that we have to at least entertain the possibility that consciousness is linked to the complex biological organization underlying life [74–81] and thus computational descriptions that capture the essence of consciousness may be much more complex than our present-day theories suggest (Figure 3). It might be impossible to 'biopsy' consciousness and remove it from its organizational dwellings. This idea contradicts many current theories of consciousness, which assume that consciousness can be captured on the abstract level of computation [47,88]. This assumption, however, might be one that will require updating in light of modern AI systems: perhaps interdependencies and organizational complexity across scales observed in living systems cannot be ignored to fully understand consciousness.

It might be that although AI systems (at least to some extent) mimic their biological counterparts on the level of network computations, in these systems we have abstracted away all the other levels of processing that causally contribute to consciousness in the living brain and, possibly, have therefore abstracted away consciousness itself. In this way, LLMs and future AI systems may be trapped in a compelling simulation of the signatures of consciousness, but without any conscious experience to speak of. If consciousness is indeed related to these other levels of processing, or their coherent interactions across scales, we might still be far from the possibility of conscious machines.

### Concluding remarks

Here, we have provided a neuroscience perspective on the possibility of consciousness in LLMs and future AI systems. We conclude that, while fascinating and alluring, LLMs are not conscious and will likely not be conscious soon. First, we detailed the vast differences between the *umwelt* of mammals (the 'slice' of the external world that they can perceive) and the highly impoverished and limited *umwelt* of LLMs when compared with biological counterparts. Second, we argue that the topological architecture of LLMs, while highly sophisticated, is sufficiently different from the neurobiological details of circuits empirically linked to consciousness in mammals that there is no *a priori* reason to conclude that they are capable of phenomenal consciousness (Figure 1). Third, we point out that it might not be possible to abstract consciousness away from the organizational complexity that is inherent within living systems but strikingly absent from AI systems. Overall, we believe that these three arguments make it extremely unlikely that LLMs, in their current

### Outstanding questions

Assessment of consciousness in LLMs and AI is often envisioned to depend on language-based tests to probe consciousness. Is it possible to evaluate consciousness based on language (i.e., text) only? If not, are we destined to remain uncertain about consciousness in LLMs, or are there any further features and aspects of consciousness that can help (more confidently) diagnose the presence (or absence) of consciousness in artificial systems?

The thalamocortical system seems to be relevant for the neural basis of consciousness in mammals. How could a thalamocortical system be implemented in AI? Which particular functions and tasks will benefit from having a thalamocortical-like system?

The ascending arousal system also plays a crucial role in facilitating consciousness in living organisms, having a complex, multifaceted role in shaping neural dynamics. To what extent does AI need to mimic these different processes to capture the computational benefits of the ascending arousal system?

Could including biological details enhance the capabilities of AI systems? Besides the thalamocortical system, dendrites seem to be key players in some of the theories of consciousness discussed in the current paper. Are dendrites just a factor that adds computational complexity/efficiency to biological neural networks, or is there more to it?

Is the organizational complexity of living systems related to consciousness? Living systems consist of various levels of processing that causally interact. Can the organizational complexity of living systems be explained more formally? New mathematical frameworks are required for dealing with such systems to shed more light on consciousness.

According to some accounts, consciousness and agency are inextricably linked. To understand how consciousness emerges from biological activity, do we first need to understand agency?

form, have the capacity for phenomenal consciousness. Rather, they mimic signatures of consciousness that are implicitly embedded within the language that people use to describe the richness of their conscious experience.

Rather than representing an antithetical account, the proposed perspective may bear some useful implications (see [Outstanding questions](#)). For one, perhaps any worries about potential moral quandaries regarding sentience in LLMs are currently more hypothetical than real ([Box 1](#)). In addition, we believe that a refined understanding of the similarities and differences in the topological architecture of LLMs and mammalian brains provides opportunities for advancing progress in both machine learning and neuroscience. To this end, major inroads will occur through mimicking features of brain organization and by learning how simple distributed systems can process elaborate information streams [89–91]. For these reasons, we are optimistic that future collaborative efforts between AI researchers and neuroscientists have the potential to gain a deeper understanding of consciousness.

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### Declaration of interests

The authors declare no competing interests in relation to this work.

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